

AI Project 4 – Optical Character Recognition (OCR)

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Project: Text Recognition using OCR

Objective

To extract text from images using Python and Tesseract OCR and convert it into editable digital text.

Technologies Used

Python 3.x, OpenCV (opencv-python), and Tesseract OCR (pytesseract).

Project Workflow

1. Read the image.
2. Convert it to grayscale.
3. Apply thresholding.
4. Extract text using Tesseract OCR.
5. Display and save the extracted text.

Code Description

The program reads an image named `sample_text.jpg`, preprocesses it using OpenCV, extracts text using `pytesseract.image_to_string()`, and saves the result into `output.txt`.

Applications

Document digitization, invoice processing, number plate recognition, passport scanning, and automatic form processing.

Future Enhancements

Real-time OCR using webcam, multilingual support, handwritten text recognition, and text-to-speech conversion.

Conclusion

This project demonstrates a simple implementation of Optical Character Recognition using Python, OpenCV, and Tesseract OCR.